



Life Cycle Cost, Eco-Efficiency, and Eco Design of Wireless Mouse

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Executive Summary

The rapid pace of technological development, paired with increasing resource consumption, presents significant sustainability challenges on a global scale. Electronics, especially modern computing devices like wireless Bluetooth mouse, contribute heavily to this issue. These products are widely used, but their short lifespans and reliance on critical, non-renewable materials make them unsustainable in the long term. This underscores the need to develop more sustainable product designs to mitigate the environmental and economic impacts of their production, use, and disposal.

This study focuses on redesigning the Bluetooth mouse to create a more sustainable alternative. It pursues three main objectives: (i) to assess the eco-efficiency of the current Bluetooth mouse by quantifying its environmental impacts using Life Cycle Assessment (LCA) and its economic impact through Life Cycle Costing (LCC); (ii) to develop alternative design options, known as eco-designs, addressing key sustainability hotspots identified in the existing product; and (iii) to identify the most sustainable design option that could serve as a viable replacement for the current model.

The analysis in this assignment revealed that the material phase of the Bluetooth mouse has the lowest eco-efficiency, primarily due to the significant quantities of minerals and other resources required for production. Often extracted through environmentally damaging processes, these materials contribute heavily to the overall environmental impact. Among the alternative designs evaluated, a solar-powered mouse demonstrated the most promising environmental performance, achieving a 19% reduction in global warming potential (GWP) compared to the base model. Other design alternatives offered minimal improvements in comparison.

While the solar-powered mouse shows potential for more sustainable outcomes, the study acknowledges several limitations. The analysis involves certain assumptions regarding manufacturing and transportation processes, which could vary significantly based on geographical or regional differences. Additionally, the study needs to consider possible future advancements in energy efficiency or improvements in sustainable material sourcing, which could alter the results. Furthermore, the end-of-life disposal processes were simplified, failing to capture the full complexity of global waste management systems.

Despite these limitations, this study offers valuable insights for brands like Logitech, providing a framework for developing more sustainable Bluetooth mouse that are both environmentally friendly and economically viable in the long run.

1.0 Introduction

As global concern for environmental sustainability intensifies, there is growing pressure on industries to design products with a reduced environmental impact. In response, Life Cycle Costing (LCC) and Life Cycle Assessment (LCA) have emerged as essential tools in product development, offering a holistic approach to evaluating both the economic and environmental impacts of a product throughout its entire lifecycle. These impacts are not limited to the production phase but extend across acquisition, use, maintenance, and final disposal. By integrating LCC and LCA, manufacturers can gain valuable insights into the long-term sustainability of their products, allowing for more informed design choices.

In the case of Bluetooth mouse, conventional design approaches often fail to account for the significant environmental burden that arises at the end of the product's life, particularly in relation to e-waste and the inadequate recycling of materials. This neglect has led to increased resource depletion and waste accumulation. To tackle these challenges, eco-design provides a promising framework that promotes sustainability at the design stage. Eco-design seeks to reduce environmental impacts by considering factors such as material selection, energy consumption, and end-of-life disposal during the initial phases of product development.

This project aims to enhance the design of the existing Bluetooth mouse by employing eco-efficiency analysis and eco-design principles, with a focus on improving its overall sustainability. The new designs for Bluetooth mouse differ significantly in terms of materials, energy consumption, and product lifespan. Modifying various components to reduce environmental impact may also affect the economic value of these designs. Therefore, eco-efficiency can serve as a crucial indicator to help identify the optimal balance between environmental and economic performance. This project aims to calculate the eco-efficiency of all proposed designs, by conducting a combined assessment of LCC and LCA results. Based on these findings, the project will suggest optimizations, enabling Logitech and other brands to design Bluetooth mouse that are both economically viable and environmentally sustainable in the future.

2.0 Life Cycle Costing (LCC) of the Bluetooth mouse

In today's competitive market, sustainability and cost-efficiency are key considerations in product design and development. The LCC approach offers a comprehensive way to assess the total cost of ownership of a product over its operational lifespan, considering both initial

purchase and ongoing operational expenses. The component of Life Cycle Cost explained in the picture below.

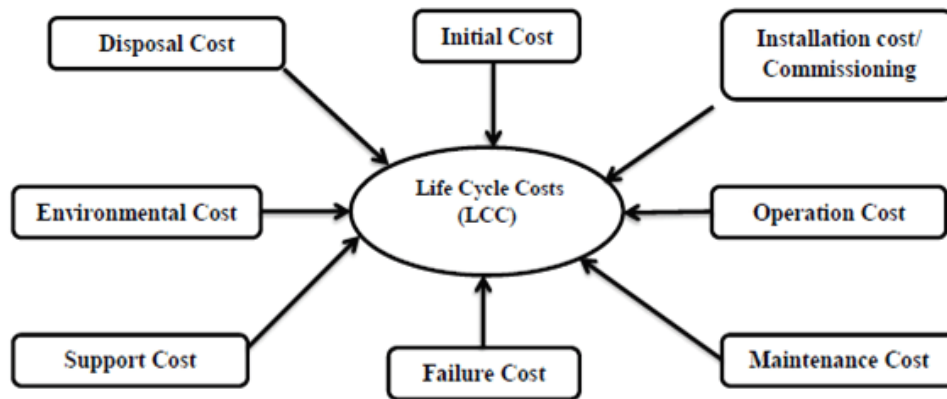


Fig. 1. Component of Life Cycle Costing

In this report, we classified the definition of Life Cycle Cost into distinct categories: material cost, production cost, transport cost, operation cost, and EOL cost. The breakdown of the options is explained in **Table 1**.

Table 1: Cost parameters for the LCC of the Bluetooth mouse

Cost segment	Value of the cost segments
Material cost	\$2.892
Production cost	\$0.130
Transport cost	\$0.231
Advertisement	\$0.976
Marketing	\$1.269
Tax	10%
Profit	40%
Product selling price	\$8.247
Battery price	\$3.20
EOL price	\$0.45
Inflation rate	3.10%
Discount rate	3.50%

2.1.1 Material Cost

Material Cost refers to the expenditure incurred on the raw materials that serve the fundamental input in the production process of the product (JSS College of Arts, Commerce and Science,

2024). In the BAU wireless bluetooth mouse, the various raw materials essential for its production, including the outer casing, electronic components, and packaging with different cost to created. Table 2 below outlines the materials used and their cost, such as ABS plastic, copper, HDPE, stainless-steel, wiring components, printed wiring board (PWB), LED Lights, printed carboard, and plastic packaging. The breakdown of the material cost explained in **Table 2**.

Table 2. Material Cost for Wireless

Input	Amount	Unit	Unit Price/kg	Value	Source
ABS plastic	39.25	g	\$13.30	\$0.522	(Chem Analyst, 2014)
Copper	0.07	g	\$9.00	\$0.001	(Southern Cross Metal Recyclres, 2024)
HDPE	0.52	g	\$11.50	\$0.006	(Xometry, 2024)
Stainless steel	1.76	g	\$11.68	\$0.021	(Melbourne Copper, 2024)
Cable with copper wire+ PVC	0.25	g	\$27.00	\$0.0068	(Renmin, 2024)
Printed wiring Board (PWB)	24	cm ²	\$850.00	\$2.04	(Renmin, 2024)
LED Light (IR movement detector)	0.22	g	\$1,000.0	\$0.22	(Digi-Key, 2024)
Printed Cardboard	6	g	\$7.00	\$0.042	(National Storage, 2024)
Plastic packaging	0.69	g	\$50.00	\$0.035	(Amazon, 2024)

2.1.2 Production Cost

Production cost encompasses all the expense directly associated with trasnforming raw materials into finished product, included materials, labor, and energy consumption also factory overhead and maintenance (Kádárová, J. 2015). For this case, production costs for manufacturing a BAU wireless bluetooth mouse is broken down into various processes, sych as molding and wire making, alongside utility usage during assembly. Each production step is calculated to reflect necessary expenditures involved in creating mouse. **Table 3** provides production cost in BAU wireless mouse.

Table 3. Production cost for Wireless Bluetooth Mouse

Input	Amount	Unit	Unit Price/kg	Value	Source
ABS plastic Moulding	39.25	g	\$3.15	\$3.15	(Rex Plastic, 2024)

Input	Amount	Unit	Unit Price/kg	Value	Source
Copper wire making	0.07	g	\$9.00	\$9.00	(Rex Plastic, 2024)
HDPE Moulding	0.52	g	\$1.30	\$1.30	(Rex Plastic, 2024)
Stainless steel wire making	1.76	g	\$3.00	\$3.00	(Rex Plastic, 2024)
Electricity (assembly)	0.06	kWh	\$1.50	\$1.50	(Citi Power, 2024)
Water	0.01	KG	\$1	\$1	(General Price, 2024)
Labor cost			\$1	\$1.20	(General Price, 2024)

2.1.3 Transport cost

Transportation costs encompass all expenses related to the movement of raw materials, after production, and finished products, including freight charges, fuel expenses, vehicle maintenance, and labor associated with logistics. The transportation costs for moving materials and finished products include various charges based on distance and weight. For instance, transporting steel wire from Wuhan to Shanghai costs \$0.001 for 1.48 kg-km, while copper wire costs only \$0.001 for 0.06 kg-km (Xinhua, 2024). Additional costs are incurred for cardboard and plastic, as well as for moving the final product from the factory to the port, and then internationally from Shanghai to Melbourne. Each leg of the journey contributes to the total transportation expense, impacting overall production costs. For the complete list transport cost can be found in **Table 4**.

Table 4. Transport cost for wireless Bluetooth mouse [source: (Freight Finder, 2024)]

Input	Amount	Unit	Unit Price/kg	Value
Steel wire _Wuhan to shanghai	1.48	KG-KM	\$0.001	\$0.001
Copper wire _Wuhan to shanghai	0.06	KG-KM	\$0.001	\$0.0001
Cardboard+Plastic _Shnghai-Factory	0.33	KG-KM	\$0.001	\$0.0001
Final product _Factory -Port	3.64	KG-KM	\$0.001	\$0.004
Final product _Shanghai-Melbourne	580.7	KG-KM	\$0.0003	\$0.174
Final product _Port -stores	4.37	KG-KM	\$0.010	\$0.044
Final product _Store-house	0.36	KG-KM	\$0.010	\$0.004
Final product _ Store-house	0.35	KG-KM	\$0.010	\$0.004

2.1.4 Marketing & advertisement cost

Understanding the upfront costs of launching a marketing campaign is crucial for effective budgeting. This initial investment encompasses expenses related to creative development,

media buying, production, and distribution. A thorough assessment of these costs helps ensure that the campaign is adequately funded and strategically planned, facilitating successful execution and measurable outcomes. For this cost based on fastercapital, the marketing and advertisement cost is 30% from total material, production, and transport cost. This indicates that the marketing costs attributed to the product or service total \$1.336.

2.1.5 Tax and profit margins

Taxes impact the overall profitability by reducing net income, while profit margins indicate how efficiently a company converts revenue into profit. To maintain healthy margins, businesses must carefully analyze their costs and pricing strategies while ensuring compliance with tax regulations. Therefore, this study assumes that the tax to enter in the Australian market is 15% and the profit margin of the Logitech is 40% on the total value of the product.

2.1.6 Usage Cost

The usage cost of a battery-operated mouse mainly involves the expenses related to battery replacements and energy consumption. Depending on the battery type such as AA, AAA, or rechargeable the costs can vary considerably. For instance, standard disposable batteries require regular replacements, while rechargeable batteries incur charging costs. If the mouse comes with a built-in rechargeable battery, electricity used for charging also be considered. As a reference, battery cost is approximately \$3.20, according to current prices on Amazon.

2.1.7 Disposal cost

End-of-Life (EOL) cost (presented in **Table 5**) refers to the expenses incurred when a product reaches the end of its useful life. This includes costs associated with disposal, recycling, or refurbishment, as well as any potential environmental fees or compliance with regulations.

Table 5: EoL cost for wireless Bluetooth mouse

Input	Amount	Unit	Unit Price/kg	Value
Consumer electronics waste management	141.76	g	\$1.54	\$0.22
Landfill (consumer electronic waste) (70%)	99.23	g	\$2.34	\$0.23

2.1.7 Inflation and Discount rate

In the March 2024 quarter, Australia's inflation for electronics and related goods, particularly in the broader category of household equipment, experienced notable deflation. As of this

period, the overall annual inflation for goods decreased to 3.5%, a sharp drop from the 9.6% peak in September 2022 (Australian Bureau of Statistics, 2024). As at the end of July 2024, the nominal local government discount rate is 3.5%

2.2 Life cycle costing result of the current product

To calculate LCC in a step-by-step approach, identify all the relevant costs over the asset's lifetime. Determine the initial acquisition cost, which includes all expenses incurred to purchase and install the asset is important. This is usually a one-time cost incurred at the start of the asset's life. Then, estimate the operational or usage costs, which represent the ongoing expenses for using the asset, such as energy, fuel, or consumables. These costs are typically calculated for each year of the asset's life and can increase due to factors like inflation. The third step is to include any maintenance costs, which account for the upkeep or repair of the asset during its operation. Maintenance costs may vary by year, depending on the complexity of the asset and the wear it experiences. Afterward, add the disposal costs, which are incurred at the end of the asset's useful life when it needs to be decommissioned or discarded.

Once all these costs are identified, the next step is to adjust for inflation, applying an inflation rate to future costs to reflect how they might increase over time. Then, use a discount rate to calculate the Net present value (NPV) of future costs, as money spent in the future is worth less than money spent today due to the time value of money. This study employs **Eq. 1** and **Eq. 2** for the inflation-adjusted cost and NPV, accordingly.

$$I_{a,t} = P_{a,t} \times (1 + i)^t \dots \dots \dots \text{Eq. 1}$$

$$NPV = \frac{R_t}{(1 + n)^t} \dots \dots \dots \text{Eq. 2}$$

Here, $I_{a,t}$ is the inflation adjusted price for the cost segment a at time t ; $P_{a,t}$ is the price for the cost segment a at time t ; i is the inflation rate; NPV is the Net Present Value; R_t is the net cash flow at time t ; n is the discount rate; t is the time of the cash flow.

Using these equations, the total NPV of the studied product at its end of life cycle is calculated which summarized in the **Table 6**.

Table 6. Calculated life cycle costing of the wireless Bluetooth mouse

Year	0	1	2	3
Initial Cost	\$10.73	\$0.00	\$0.00	\$0.00
Usage	\$3.20	\$3.30	\$3.40	\$0.00

2.3.2 Analysis of the eco-efficiency of the product

It is evident from **Fig. 2** (also summarized in **Table A.2**) that the material phase has the lowest ecoefficiency than other life cycle phases of the existing product. It is because it requires a lot of minerals and materials which include the impact of the extractions too. At the same time, the material prices are comparatively cheaper due to their availability.

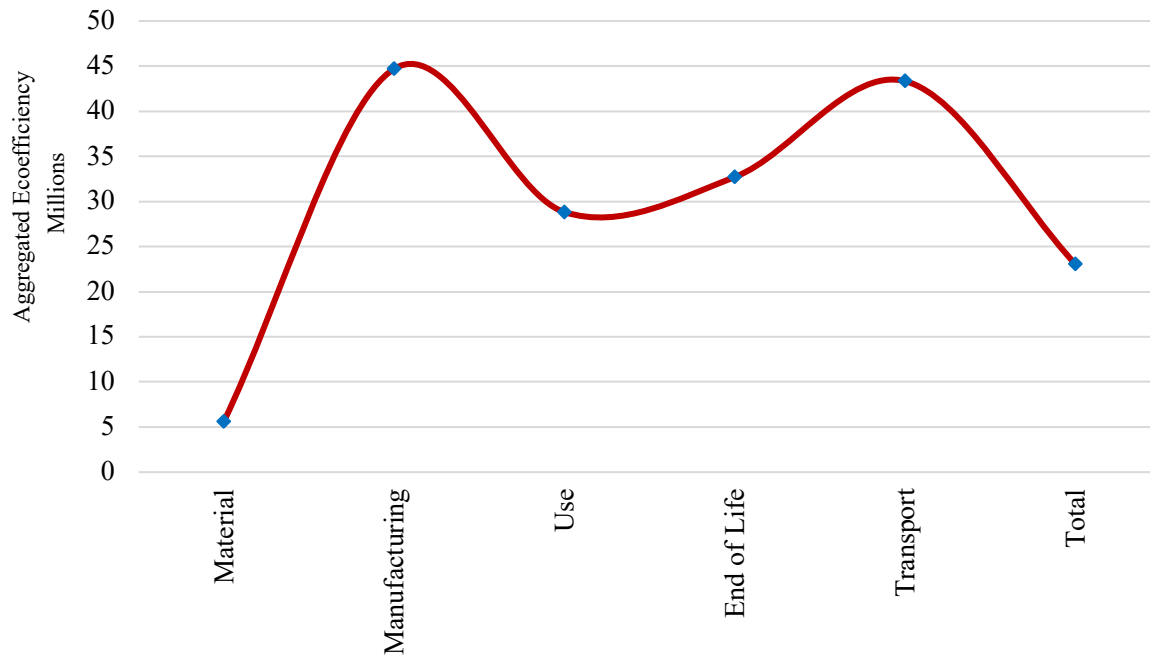


Fig 2. Aggregated ecoefficiency of the Bluetooth mouse

The second most impactful phase with the lower ecoefficiency is the use phase which is quite higher than the material phase (i.e., ~413% higher). It is entirely because of the Alkaline battery. The manufacturing process and the materials used in the battery have higher environmental impacts and thus using those batteries makes the use phase less eco-efficient. Therefore, it is evident that the design modification in the material and use phase can increase the overall sustainability of the product.

3.0 Eco design proposal

Based on the comprehensive analysis and the findings from **section 2.0**, three alternative design options (presented in **Fig. 3**) were selected for comparison with the current model. They are as follows:

- i) Solar-powered mouse that integrates photovoltaic (PV) cells into the mouse, allowing it to be powered by solar energy. By eliminating the need for disposable

alkaline batteries during the use phase, this design significantly reduces environmental impacts associated with battery production, usage, and disposal.

- ii) Mouse with recycled plastic content to address the environmental concerns related to the production of virgin ABS plastic. By reducing the demand for new plastic in the material phase, the ecological footprint of the mouse can be minimized, contributing to resource conservation.
- iii) Mouse with an extended lifespan. This design focuses on increasing the durability and lifespan of the mouse, thereby reducing the frequency of replacements and the associated environmental impacts across its life cycle. By extending the product's usable life, fewer resources are consumed, and waste generation is minimized over time.

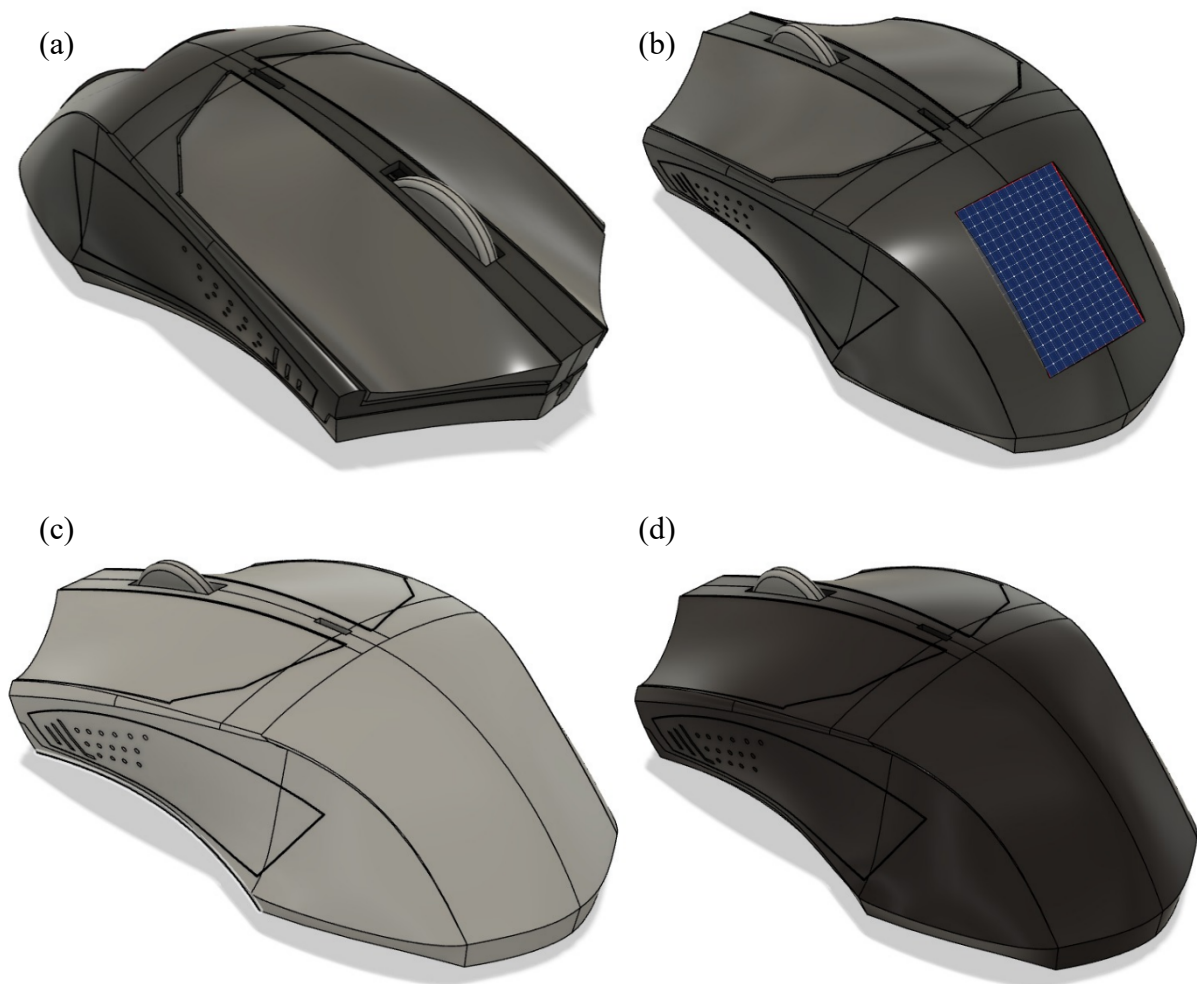


Fig. 3: Proposed designed option for the Bluetooth mouse a) BAU- Business-as-usual or the regular mouse (Logitech), b) Option 1- solar-powered mouse, c) Option 2- mouse with the recycled ABS plastics, c) Option 3- mouse with the extended life cycle (i.e., 6 years of lifetime)

5.0 Life cycle analysis (LCA) of the proposed designs

LCA, guided by the ISO 14040 and ISO 14044 standards, is a comprehensive and widely recognized tool for assessing the environmental impacts of products and processes throughout their entire life cycle, from raw material extraction to disposal (Finkbeiner et al., 2006). It examines products from a life cycle perspective and considers a wide array of environmental impacts (Finnveden and Potting, 2014). A traditional LCA has four stages, such as (i) Goal and scope definition, (ii) Life Cycle Inventory (LCI), (iii) Life Cycle Impact Assessment (LCIA), and (iv) Interpretation (Muralikrishna and Manickam, 2017). This project employs the LCA tool using ISO 14040 methodology and covering the entire lifecycle from raw material extraction through production, usage, and end-of-life disposal of a computer mouse. For that, the Open LCA 2.2 and the EcoInvent 3.7 database were employed in this project. In a typical office environment, a computer mouse is used daily for an 8-hour workday, with an average of around 400 clicks per day. With proper maintenance, a computer mouse can have an average lifespan of 3 years (MeeTion, 2024). Therefore, the functional unit defined for this project is “a wireless Bluetooth mouse to perform average 400 clicks daily over a period of 3 years.”

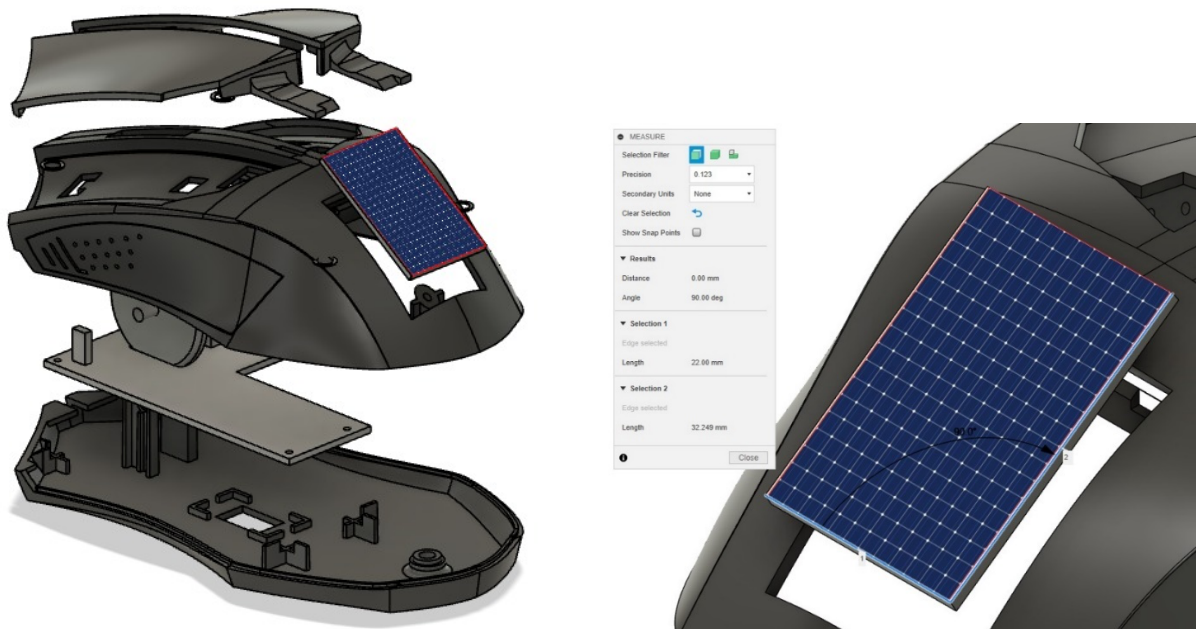


Fig. 4: Solar panel adjusted modified design of the mouse (a) exploded view of the product, (b) PV panel measurement

5.1 Life cycle inventory development

The LCI for computer mouse designs involves the systematic collection of data on the inputs and outputs associated with each stage of its life cycle, from raw material extraction, manufacturing, and assembly to its use, transportation, and eventual disposal. This data includes the consumption of resources like energy and materials, providing a detailed foundation for assessing the environmental impacts of the computer mouse. Besides, the measurement of the solar PV cells in the mouse (for option 1) was derived from the 3D modeling, which is presented in **Fig. 4**. Besides, **Table 7** summarizes the detailed LCI of all the proposed design options including the existing Bluetooth mouse.

Table 7: Life cycle inventory (LCI) of the different design options

Phase	Input	BAU	Option 1	Option 2	Option 3	Unit	Ecoinvent background data
Material	ABS plastic	39.25	39.25	31.4	19.63	g	market for acrylonitrile-butadiene-styrene copolymer acrylonitrile-butadiene-styrene copolymer Cutoff, S - GLO
	Copper	0.07	0.07	0.07	0.04	g	market for copper, cathode copper, cathode Cutoff, S - GLO
	HDPE	0.52	0.52	0.52	0.26	g	market for polyethylene, high density, granulate polyethylene, high density, granulate Cutoff, S - GLO
	Stainless steel	1.76	1.76	1.76	0.88	g	market for iron-nickel-chromium alloy iron-nickel-chromium alloy Cutoff, S - GLO
	Cable with copper wire+ PVC	0.25	0.25	0.25	0.13	g	market for cable, unspecified cable, unspecified Cutoff, S - GLO
	Printed wiring Board (PWB) (5.5 gram)	24	24	24	12.00	cm ²	market for printed wiring board, for surface mounting, Pb containing surface printed wiring board, for surface mounting, Pb containing surface Cutoff, S - GLO
	LED Light (IR movement detector)	0.22	0.22	0.22	0.11	g	market for light emitting diode light emitting diode Cutoff, S - GLO

Phase	Input	BAU	Option 1	Option 2	Option 3	Unit	Ecoinvent background data
	Printed Cardboard	6	6	6	3.00	g	market for folding boxboard carton folding boxboard carton Cutoff, S - RoW
	Plastic packaging	0.69	0.69	0.69	0.35	g	market for packaging film, low density polyethylene packaging film, low density polyethylene Cutoff, S - GLO
	Solar panel		0.0009		0.00		market for photovoltaic cell, multi-Si wafer photovoltaic cell, multi-Si wafer Cutoff, S - GLO
	ABS plastic Moulding	39.25	39.25	39.25	19.63	g	market for injection moulding injection moulding Cutoff, S - GLO
	Copper wire making	0.07	0.07	0.07	0.04	g	market for wire drawing, copper wire drawing, copper Cutoff, S - GLO
Manufac turing/ Assembl ing	HDPE Moulding	0.52	0.52	0.52	0.26	g	market for injection moulding injection moulding Cutoff, S - GLO
	Stainless steel wire making	1.76	1.76	1.76	0.88	g	market for wire drawing, steel wire drawing, steel Cutoff, S - GLO
	Electricity (assembly)	0.06	0.06	0.06	0.03	kWh	electricity voltage transformation from high to medium voltage electricity, medium voltage APOS, S - CN-CSG
Use	Water	0.01	0.01	0.01	0.01	KG	market for tap water tap water Cutoff, S - RoW
	AA battery (1.5 V)	69		69	115	g	market for battery, NaCl battery, NaCl Cutoff, S - GLO
	Consumer electronics waste management	141.8	141.76	141.76	70.88	g	treatment of waste plastic, consumer electronics, dismantling, sorting and cleaning, formal sector waste plastic, consumer electronics, sorted Cutoff, S - IN
EoL	Landfill (consumer electronic waste) (70%)	99.23	99.23	99.23	49.62	g	treatment of waste plastic, consumer electronics, sanitary landfill, wet infiltration class (500mm) waste plastic, consumer electronics Cutoff, S - GLO
Transpo rt	Steel wire _Wuhan to shanghai	1.48	1.48	1.48	0.74	KG-KM	market group for transport, freight, lorry, unspecified transport, freight, lorry, unspecified Cutoff, S - GLO

Phase	Input	BAU	Option 1	Option 2	Option 3	Unit	Ecoinvent background data
	Copper wire						
	_Wuhan to	0.06	0.06	0.06	0.03	KG-	market group for transport, freight,
	shanghai					KM	lorry, unspecified transport, freight,
	Cardboard+Plast						lorry, unspecified Cutoff, S - GLO
	ic_Shnghai-	0.33	0.33	0.33	0.17	KG-	market group for transport, freight,
	Factory					KM	lorry, unspecified transport, freight,
	Final						lorry, unspecified Cutoff, S - GLO
	product_Factory	3.64	3.64	3.64	1.82	KG-	market group for transport, freight,
	-Port					KM	lorry, unspecified transport, freight,
	Final						lorry, unspecified Cutoff, S - GLO
	product_Shangh	580.7	580.7	580.7	290.35	KG-	market for transport, freight, sea,
	ai-Melbourne					KM	container ship transport, freight, sea,
	Final						container ship Cutoff, S - GLO
	product_Port -	4.37	4.37	4.37	2.19	KG-	market for transport, freight, light
	stores					KM	commercial vehicle, EURO1 transport,
	Final product_	0.36	0.36	0.36	0.18	KG-	freight, light commercial vehicle,
	Store-house					KM	EURO1 Cutoff, S - ZA
	Final product_	0.35	0.35	0.35	0.18	KG-	market for transport, freight, light
	Store-house					KM	commercial vehicle, EURO1 transport,
							freight, light commercial vehicle,
							EURO1 Cutoff, S - ZA

5.2 Life cycle impact assessment

For the LCIA, this study employed the ReCiPe 2016 midpoint impact assessment method. The ReCiPe 2016 midpoint impact assessment method translates LCI data into environmental impacts at a midpoint level, focusing on specific environmental categories such as climate change, ozone depletion, resource depletion, etc (Huijbregts et al., 2016).

Table 8 (a) and **Table 8 (b)** show the environmental impacts of all the design options. The tables also present the impacts based on the life cycle phases of the proposed designs. All the impact categories are compiled in the tables.

Table 8 (a): Environmental impacts of all the design proposals

Name	Unit	Material				Manufacturing/Assembly				Use			
		BAU	Option 1	Option 2	Option 3	BAU	Option 1	Option 2	Option 3	BAU	Option 1	Option 2	Option 3
Fine particulate matter formation	kg PM2.5 eq	2.43E-03	2.72E-03	2.40E-03	1.21E-03	1.62E-04	1.62E-04	1.62E-04	8.11E-05	6.15E-03		6.15E-03	1.03E-02
Fossil resource scarcity	kg oil eq	3.15E-01	3.55E-01	2.98E-01	1.57E-01	2.45E-02	2.45E-02	2.45E-02	1.23E-02	1.35E-01		1.35E-01	2.24E-01
Freshwater ecotoxicity	kg 1,4-DCB	3.18E-01	3.29E-01	3.18E-01	1.59E-01	1.81E-03	1.81E-03	1.81E-03	9.05E-04	3.14E-01		3.14E-01	5.23E-01
Freshwater eutrophication	kg P eq	8.58E-04	9.36E-04	8.56E-04	4.29E-04	3.05E-05	3.05E-05	3.05E-05	1.53E-05	5.01E-04		5.01E-04	8.34E-04
Global warming	kg CO2 eq	1.12E+00	1.27E+00	1.09E+00	5.62E-01	9.40E-02	9.40E-02	9.40E-02	4.70E-02	4.86E-01		4.86E-01	8.11E-01
Human carcinogenic toxicity	kg 1,4-DCB	9.97E-02	1.17E-01	9.84E-02	4.99E-02	4.67E-03	4.67E-03	4.67E-03	2.33E-03	2.37E-01		2.37E-01	3.96E-01
Human non-carcinogenic toxicity	kg 1,4-DCB	3.28E+00	3.51E+00	3.28E+00	1.64E+00	3.33E-02	3.33E-02	3.33E-02	1.66E-02	3.23E+00		3.23E+00	5.39E+00
Ionizing radiation	kBq Co-60 eq	9.20E-02	1.06E-01	9.19E-02	4.60E-02	9.18E-03	9.18E-03	9.18E-03	4.59E-03	9.18E-02		9.18E-02	1.53E-01
Land use	m2a crop eq	2.88E-02	3.22E-02	2.88E-02	1.44E-02	3.31E-03	3.31E-03	3.31E-03	1.65E-03	2.17E-02		2.17E-02	3.61E-02
Marine ecotoxicity	kg 1,4-DCB	4.04E-01	4.23E-01	4.04E-01	2.02E-01	2.44E-03	2.44E-03	2.44E-03	1.22E-03	4.04E-01		4.04E-01	6.74E-01
Marine eutrophication	kg N eq	5.73E-05	7.26E-05	5.63E-05	2.87E-05	2.35E-06	2.35E-06	2.35E-06	1.18E-06	5.24E-05		5.24E-05	8.73E-05
Mineral resource scarcity	kg Cu eq	4.39E-04	4.59E-04	4.30E-04	2.19E-04	1.13E-05	1.13E-05	1.13E-05	5.64E-06	2.65E-03		2.65E-03	4.41E-03
Ozone formation, Human health	kg NOx eq	2.68E-03	3.04E-03	2.62E-03	1.34E-03	2.26E-04	2.26E-04	2.26E-04	1.13E-04	2.02E-03		2.02E-03	3.36E-03
Ozone formation, Terrestrial ecosystems	kg NOx eq	2.75E-03	3.14E-03	2.68E-03	1.37E-03	2.30E-04	2.30E-04	2.30E-04	1.15E-04	2.06E-03		2.06E-03	3.43E-03
Stratospheric ozone depletion	kg CFC11 eq	5.20E-07	5.90E-07	5.18E-07	2.60E-07	3.03E-08	3.03E-08	3.03E-08	1.51E-08	3.38E-07		3.38E-07	5.64E-07
Terrestrial acidification	kg SO2 eq	4.84E-03	5.33E-03	4.75E-03	2.42E-03	3.22E-04	3.22E-04	3.22E-04	1.61E-04	1.99E-02		1.99E-02	3.32E-02
Terrestrial ecotoxicity	kg 1,4-DCB	1.91E+01	2.40E+01	1.91E+01	9.57E+00	5.64E-02	5.64E-02	5.64E-02	2.82E-02	2.54E+01		2.54E+01	4.23E+01
Water consumption	m3	6.29E+00	1.01E+01	6.28E+00	3.14E+00	1.40E+00	1.40E+00	1.40E+00	6.98E-01	6.42E+00		6.42E+00	1.07E+01

Table 8 (b): Environmental impacts of all the design proposals

Name	Unit	End of Life				Transport			
		BAU	Option 1	Option 2	Option 3	BAU	Option 1	Option 2	Option 3
Fine particulate matter formation	kg PM2.5 eq	1.59E-05	1.59E-05	1.59E-05	7.94E-06	4.18E-05	4.18E-05	4.18E-05	2.09E-05
Fossil resource scarcity	kg oil eq	2.63E-03	2.63E-03	2.63E-03	1.32E-03	2.63E-03	2.63E-03	2.63E-03	1.32E-03
Freshwater ecotoxicity	kg 1,4-DCB	8.96E-02	8.96E-02	8.96E-02	4.48E-02	2.38E-04	2.38E-04	2.38E-04	1.19E-04
Freshwater eutrophication	kg P eq	6.41E-06	6.41E-06	6.41E-06	3.21E-06	1.51E-06	1.51E-06	1.51E-06	7.55E-07
Global warming	kg CO2 eq	3.55E-02	3.55E-02	3.55E-02	1.78E-02	8.98E-03	8.98E-03	8.98E-03	4.49E-03
Human carcinogenic toxicity	kg 1,4-DCB	2.76E-03	2.76E-03	2.76E-03	1.38E-03	8.26E-04	8.26E-04	8.26E-04	4.13E-04
Human non-carcinogenic toxicity	kg 1,4-DCB	3.96E-01	3.96E-01	3.96E-01	1.98E-01	4.37E-03	4.37E-03	4.37E-03	2.19E-03
Ionizing radiation	kBq Co-60 eq	2.16E-04	2.16E-04	2.16E-04	1.08E-04	1.48E-04	1.48E-04	1.48E-04	7.39E-05
Land use	m2a crop eq	4.33E-04	4.33E-04	4.33E-04	2.16E-04	1.21E-04	1.21E-04	1.21E-04	6.06E-05
Marine ecotoxicity	kg 1,4-DCB	1.13E-01	1.13E-01	1.13E-01	5.63E-02	3.29E-04	3.29E-04	3.29E-04	1.65E-04
Marine eutrophication	kg N eq	1.30E-04	1.30E-04	1.30E-04	6.52E-05	1.15E-07	1.15E-07	1.15E-07	5.76E-08
Mineral resource scarcity	kg Cu eq	6.88E-07	6.88E-07	6.88E-07	3.44E-07	5.17E-07	5.17E-07	5.17E-07	2.58E-07
Ozone formation, Human health	kg NOx eq	3.69E-05	3.69E-05	3.69E-05	1.85E-05	1.27E-04	1.27E-04	1.27E-04	6.37E-05
Ozone formation, Terrestrial ecosystems	kg NOx eq	3.77E-05	3.77E-05	3.77E-05	1.89E-05	1.29E-04	1.29E-04	1.29E-04	6.43E-05
Stratospheric ozone depletion	kg CFC11 eq	1.54E-08	1.54E-08	1.54E-08	7.70E-09	5.67E-09	5.67E-09	5.67E-09	2.84E-09
Terrestrial acidification	kg SO2 eq	2.86E-05	2.86E-05	2.86E-05	1.43E-05	1.27E-04	1.27E-04	1.27E-04	6.33E-05
Terrestrial ecotoxicity	kg 1,4-DCB	2.23E-02	2.23E-02	2.23E-02	1.11E-02	2.09E-02	2.09E-02	2.09E-02	1.05E-02
Water consumption	m3	1.41E-02	1.41E-02	1.41E-02	7.07E-03	7.22E-03	7.22E-03	7.22E-03	3.61E-03

6.0 Comparative assessment based on LCA results

Fig. 5 provides a comprehensive overview of the environmental impacts across all midpoint impact categories, with each design option represented by distinct colors for clarity. Among the options, Option 1 consistently performs the best, showing the lowest environmental impacts in almost every category. A key factor contributing to the higher impacts in the BAU option is the use of alkaline batteries during the use phase. Alkaline batteries are known for their significant environmental burden, which magnifies the overall impact of the BAU option.

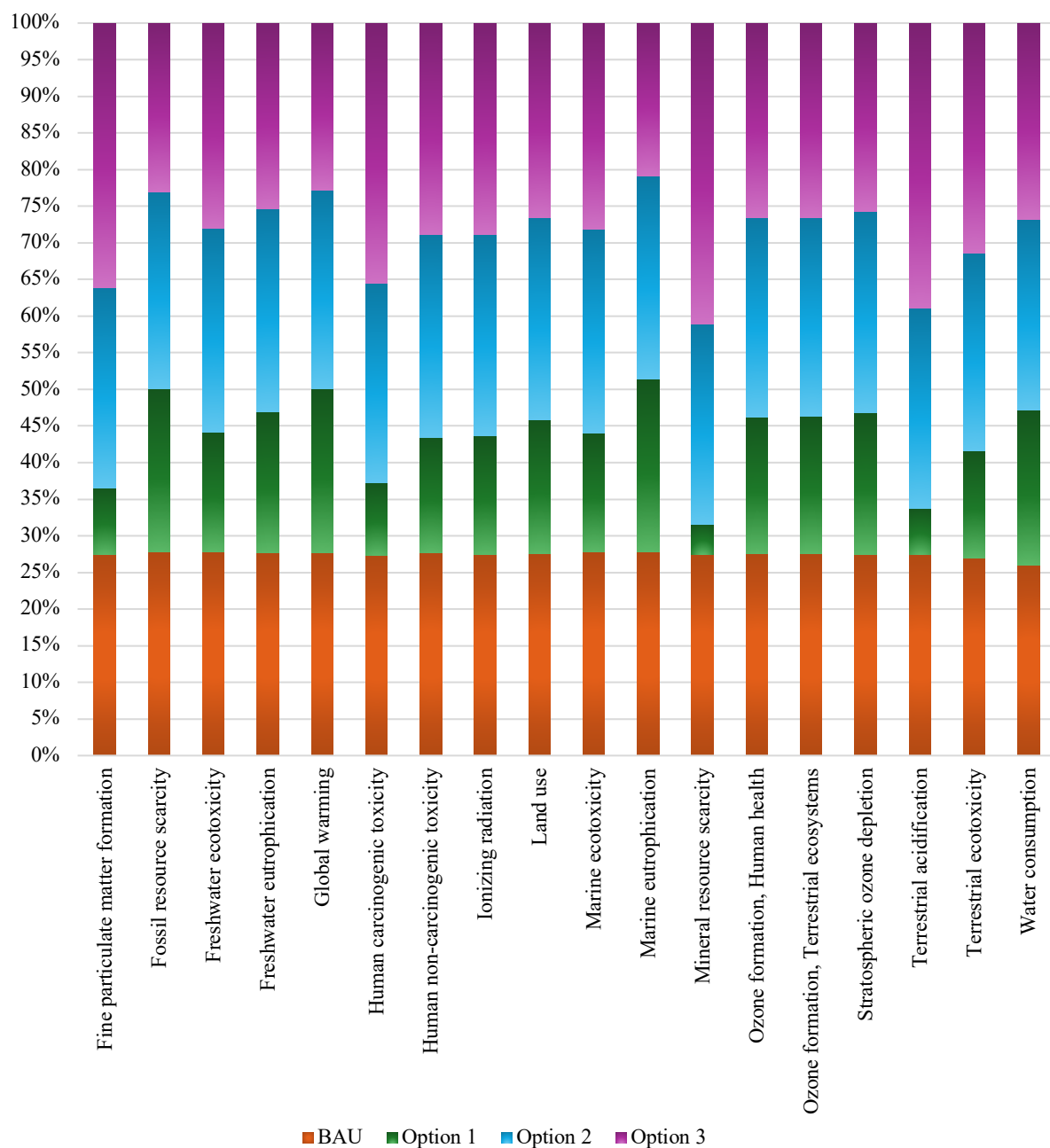


Fig. 5: Comparative assessment of different design options with the current product (BAU)

In contrast, Option 1 eliminates the use of alkaline batteries by incorporating solar panels into the design. This shift plays a crucial role in substantially reducing the environmental footprint of Option 1, particularly in categories such as resource depletion and energy use. The renewable energy contribution from solar panels effectively reduces dependency on traditional battery technologies, resulting in a more sustainable outcome.

However, it's worth noting that the reduction in global warming potential, while present, is not as pronounced as in other impact categories. This is largely because the production of solar PV panels involves the extraction and processing of rare earth materials, which are associated with significant environmental harm. The mining and refining of these materials are resource-intensive processes that contribute to greenhouse gas emissions, partially offsetting the benefits of using solar energy. Despite this, the overall environmental performance of Option 1 remains superior, with solar integration offering long-term sustainability benefits in many areas.

The comparative analysis of global warming potential across the four design options (**Fig. 6**) shows that Option 1 performs the best, with the lowest emissions at 1.412 kg CO₂ eq, representing a 19% reduction compared to the BAU option, which has the highest emissions at 1.748 kg CO₂ eq. This reduction is likely due to the integration of solar panels, eliminating the need for alkaline batteries in the use phase. Besides, Option 2 and Option 3 also offer reductions, but to a lesser extent. Option 2 has a global warming potential of 1.710 kg CO₂ eq, making it slightly better than BAU but significantly less effective than Option 1. Option 3 performs somewhat better at 1.441 kg CO₂ eq, closer to Option 1 but still not as impactful in reducing emissions. Overall, Option 1 is clearly the most environmentally sustainable choice in terms of global warming potential, while Options 2 and 3 show only moderate improvements.

Fig. 6 (b) reveals that Option 1 significantly reduces global warming potential during the use phase, achieving a 100% reduction in emissions compared to the BAU and Option 2, which maintain high emissions in this phase. However, Option 1 shows a 13% increase in material-phase emissions due to the use of solar panels. In contrast, Option 3 reduces material-phase emissions by nearly 50%, though it has the highest use-phase emissions, nearly 67% higher than BAU because of using more alkaline batteries in this phase. Manufacturing emissions remain consistent for BAU, Option 1, and Option 2, while Option 3 cuts these emissions by 50%. Transport and end-of-life phase impacts are relatively minor, with Option 3 showing the lowest emissions in these phases. Therefore, it is clearly evident that Option 1 performs the best in reducing use-phase emissions, while Option 3 excels in material, manufacturing, and transport phases.

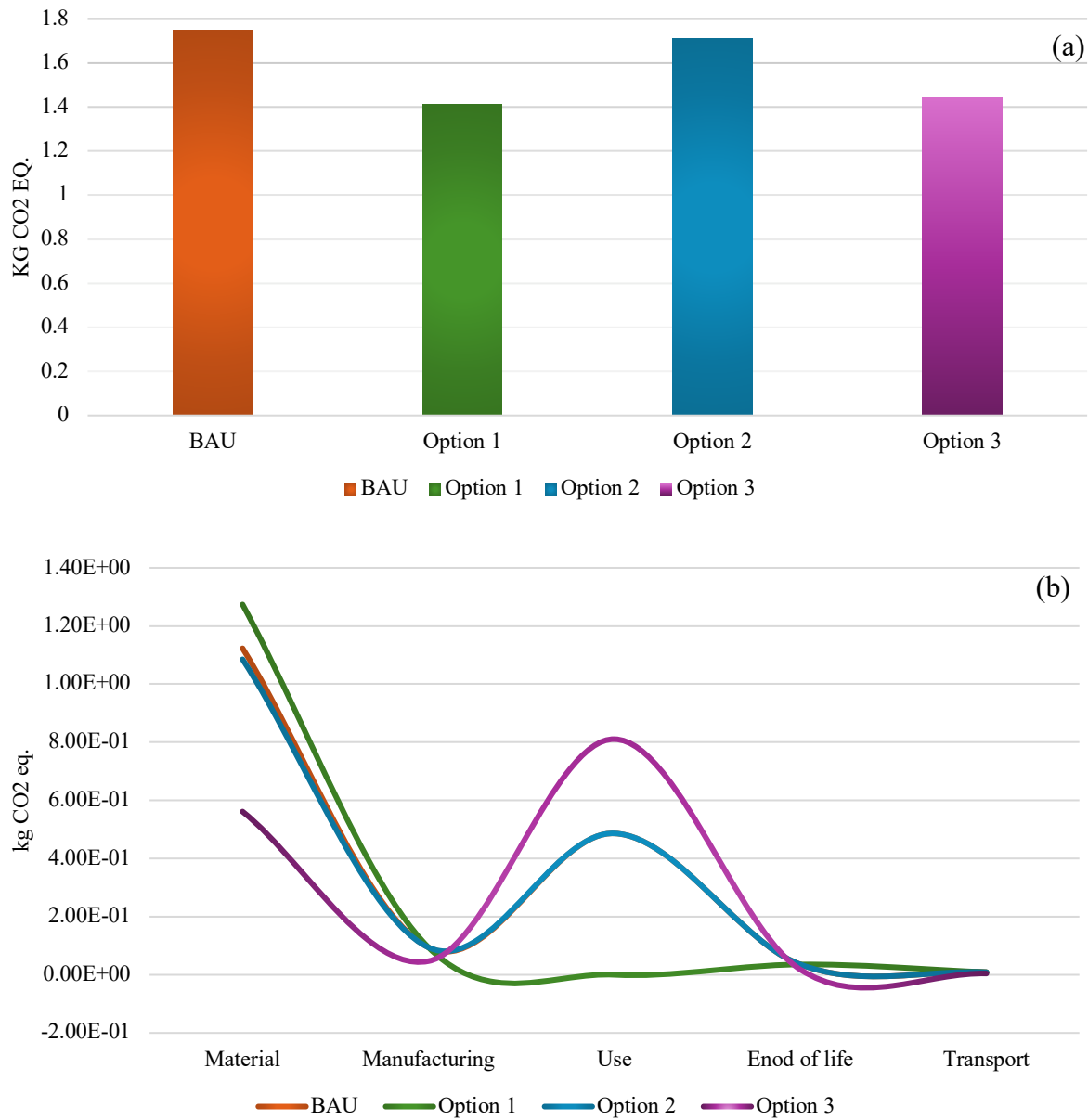


Fig. 6: Global warming potential of different design options (a) total impact, (b) phase-wise impacts

6.0 Discussion

The eco-design comparison revealed that the solar-powered mouse is the most sustainable option among the models analyzed. This design demonstrated the lowest environmental impact, particularly in reducing global warming potential, as it significantly decreases energy consumption during the product's use phase. By utilizing solar energy as the primary power source, this option eliminates the need for disposable batteries, reducing electronic waste while

also minimizing the resource extraction and energy required for battery production and disposal.

Logitech, the brand featured in this analysis, can leverage these findings by incorporating solar-powered technology into its future product lines. This strategy aligns well with Logitech's ongoing sustainability initiatives, reinforcing its reputation as a forward-thinking, eco-conscious company. By adopting solar-powered designs, Logitech can offer consumers a greener alternative, addressing the increasing demand for sustainable products. Furthermore, this innovation could serve as a differentiator in the competitive market, attracting environmentally conscious consumers and unlocking new market opportunities.

Beyond the solar-powered option, Logitech can further enhance environmental sustainability by integrating features from the other eco-designs analyzed. For example, using recycled plastics in the mouse casing would reduce the need for virgin materials, thereby lowering the carbon footprint associated with manufacturing. This approach can be scaled effectively without compromising product durability, offering an additional sustainability benefit. Furthermore, incorporating design elements from the extended-life mouse would enhance product longevity, reducing the frequency of replacements and minimizing resource consumption over time.

By combining the key aspects of these three eco-designs—solar power for energy efficiency, recycled materials to reduce resource extraction, and extended product life for durability—Logitech can create a comprehensive solution to significantly minimize the environmental impact of its products. This hybrid approach would not only reduce the company's carbon footprint but also provide customers with high-performance, eco-friendly products.

6.4 Conclusion

Our current study shows that the success of eco-design depends on considering all stages of the product life cycle in an integrated manner and optimising them in relation to environmental and economic benefits. By integrating LCC and LCA approaches, companies can make more sustainable design decisions that not only improve on environmental benefits, but also balance long-term economics. This provides valuable practical guidance for the design of future Bluetooth mouse and other consumer electronics products, especially in the context of the global drive for sustainability.

7.0 Reflection

The completion of these two assignments has provided a comprehensive understanding of Life Cycle Assessment (LCA) and Life Cycle Costing (LCC), as well as their application in evaluating product sustainability. The first assignment focused on LCA, which allowed us to assess the environmental impacts of a product throughout its entire lifecycle, from raw material extraction to disposal. This method emphasized the importance of understanding the environmental footprint of each phase, particularly the resource-intensive processes such as material extraction and manufacturing. Besides, the second assignment introduced Life Cycle Costing (LCC), along with eco-efficiency and eco-design principles. LCC provided a way to evaluate the total cost associated with a product's lifecycle, while eco-efficiency helped balance both environmental and economic considerations. Following the LCC result, a base model computer mouse is then compared with alternative eco-design options, such as a solar-powered mouse, a recycled plastic mouse, and a mouse designed for extended life. The comparison highlighted how eco-design choices, in general, can significantly improve environmental outcomes by minimizing material usage, reducing emissions, and extending product life cycles.

In general, these assignments highlighted the importance of integrating both LCA and LCC when assessing product sustainability. LCA allowed a detailed analysis of environmental impacts, whereas LCC ensured that economic factors were equally considered. Eco-design principles further strengthened the evaluation, providing insights into how sustainable design can lead to both reduced environmental impacts and cost savings. This collaborative work provided valuable experience with tools like OpenLCA and underscored the importance of adopting a holistic approach to product development. We believe that gaining a deep understanding of sustainability measurement tools and applying them within complex systems to quantify true sustainability will greatly enhance our ability as engineers. This knowledge will enable us to design products, services, systems, economies, and societies in a more sustainable and responsible way.

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Appendix

Table A.1: Midpoint impacts of the current bluetooth mouse (extracted from assignment 1)

Name	Unit	Material	Manufacturing/ Assembly	Use	End of Life	Transport	Total
Fine particulate matter formation	kg PM2.5 eq	0.002429314	1.62E-04	0.006150645	1.59E-05	4.18E-05	0.00879977
Fossil resource scarcity	kg oil eq	0.314682482	0.024540489	0.134567306	0.002634617	0.002632715	0.47905761
Freshwater ecotoxicity	kg 1,4-DCB	0.318033446	0.001809279	0.313572234	0.089633543	2.38E-04	0.72328636
Freshwater eutrophication	kg P eq	8.58E-04	3.05E-05	5.01E-04	6.41E-06	1.51E-06	0.00139693
Global warming	kg CO2 eq	1.123097038	0.094028742	0.486408777	0.035508008	0.008984438	1.748027
Human carcinogenic toxicity	kg 1,4-DCB	0.099733788	0.004668955	0.237410845	0.002760314	8.26E-04	0.34539951
Human non-carcinogenic toxicity	kg 1,4-DCB	3.282256335	0.0332874	3.232637246	0.395715347	0.004372899	6.94826923
Ionizing radiation	kBq Co-60 eq	0.092010963	0.009182433	0.09176954	2.16E-04	1.48E-04	0.19332726
Land use	m2a crop eq	0.028816867	0.003306045	0.021675586	4.33E-04	1.21E-04	0.05435251
Marine ecotoxicity	kg 1,4-DCB	0.404473977	0.00243501	0.404153241	0.112636139	3.29E-04	0.92402751
Marine eutrophication	kg N eq	5.73E-05	2.35E-06	5.24E-05	1.30E-04	1.15E-07	0.00024264
Mineral resource scarcity	kg Cu eq	4.39E-04	1.13E-05	0.002648046	6.88E-07	5.17E-07	0.00309918
Ozone formation, Human health	kg NOx eq	0.002675537	2.26E-04	0.002015244	3.69E-05	1.27E-04	0.00508097
Ozone formation, Terrestrial ecosystems	kg NOx eq	0.002746305	2.30E-04	0.002057335	3.77E-05	1.29E-04	0.00519954
Stratospheric ozone depletion	kg CFC11 eq	5.20E-07	3.03E-08	3.38E-07	1.54E-08	5.67E-09	9.0984E-07
Terrestrial acidification	kg SO2 eq	0.004840274	3.22E-04	0.019902133	2.86E-05	1.27E-04	0.02521969
Terrestrial ecotoxicity	kg 1,4-DCB	19.13710354	0.056436653	25.39029517	0.022297269	0.020905264	44.6270379
Water consumption	m3	6.285950061	1.395030269	6.415946991	0.014142444	0.007221363	14.1182911

Table A.2 Midpoint impact category wise ecoefficiencies and the aggregated result of ecoefficiency for different life cycle phases

Life cycle phase	Material	Manufacturing/ Assembly	Use	End of Life	Transport	Total
Fine particulate matter formation	1190.46	8201.98	1573.82	30847.89	5530.37	2372.79
Fossil resource scarcity	9.19	54.20	71.93	185.99	87.74	43.59
Freshwater ecotoxicity	9.09	735.10	30.87	5.47	971.19	28.87
Freshwater eutrophication	3371.15	43556.86	19336.75	76406.00	152881.52	14947.09
Global warming	2.58	14.14	19.90	13.80	25.71	11.94
Human carcinogenic toxicity	29.00	284.86	40.77	177.52	279.79	60.45
Human non-carcinogenic toxicity	0.88	39.96	2.99	1.24	52.83	3.01
Ionizing radiation	31.43	144.84	105.48	2263.69	1562.25	108.00
Land use	100.36	402.29	446.59	1132.34	1904.61	384.16
Marine ecotoxicity	7.15	546.20	23.95	4.35	701.82	22.60
Marine eutrophication	50466.93	565764.25	184736.18	3755.54	2005278.78	86052.05
Mineral resource scarcity	6592.99	117922.43	3655.53	712130.22	446884.03	6737.27
Ozone formation, Human health	1080.90	5888.69	4803.39	13264.39	1813.28	4109.45
Ozone formation, Terrestrial ecosystems	1053.05	5791.04	4705.12	12989.99	1797.55	4015.74
Stratospheric ozone depletion	5559871.22	43947862.16	28608826.01	31836649.69	40716459.04	22949073.95
Terrestrial acidification	597.49	4128.16	486.38	17158.49	1825.41	827.92
Terrestrial ecotoxicity	0.15	23.57	0.38	21.98	11.05	0.47
Water consumption	0.46	0.95	1.51	34.65	31.99	1.48
Aggregated ecoefficiency	5624414.48	44701361.67	28828867.54	32707043.23	43338098.96	23068800.83